IPW3 : NACA0012 Code-to-Code Comparisons

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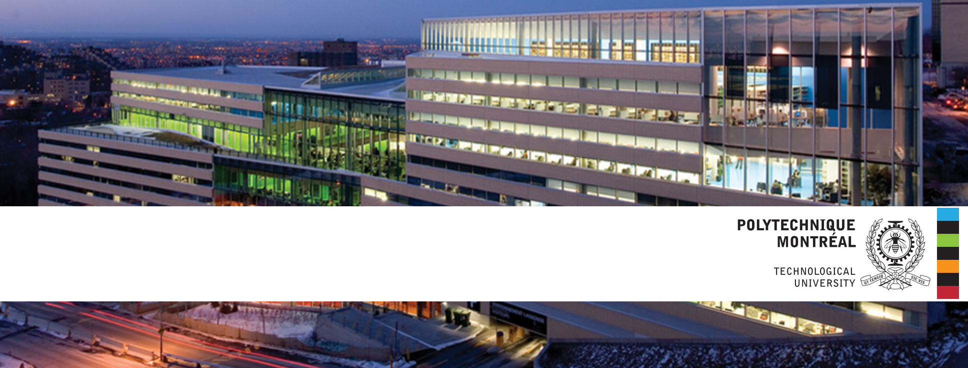

POLYTECHNIQUE MONTRÉAL
TECHNOLOGICAL UNIVERSITY

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Geometry and Aerodynamic Conditions

NASA Icing Research Tunnel

Airfoil	NACA0012
Chord, c	$0.5334\ \mathrm{m}$
Angle of attack	4°

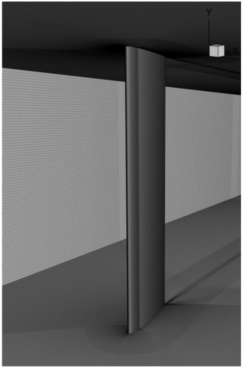
Aerodynamic Conditions

M_∞	0.3114
U_∞	$102.15\ \mathrm{m/s}$
T_∞	$267.75\ \mathrm{K}$
ρ_∞	$1.1884\ \mathrm{kg/m^3}$
Re_c	3.8441×10^6

Quantities of Interest

$$C_L, \quad C_D, \quad C_M, \quad C_p$$

Icing Conditions



Grid	Nodes	Elements
L1	60.45 M	59.86 M
L2	27.31 M	26.99 M
L3	11.23 M	11.08 M
L4	5.95 M	5.86 M

	AE3932	**AE3933**
T_0 [°C]	-5.6	-2.2
T_∞ [K]	262.305	265.8
M_∞	0.3151	0.3127
U_∞ [m/s]	102.31	102.19
LWC [g/m^3]	0.55	0.55
MVD [μm]	20	20
Exposure time [min]	7	7
Exp. ice mass [g]	**101.0**	**85.9**

Experimental Observation

The warmer AE3933 condition produces approximately **15% less ice mass** than AE3932 (85.9 g vs. 101.0 g). This difference will be revisited when assessing the predicted thermodynamic mass balance and ice accretion.

Droplet Distribution

- Single-bin to 15-bin distributions
- Total LWC preserved between distributions
- Evaluated across all four grid levels

Quantities of Interest

$$\beta, \quad HTC, \quad T_s, \quad f, \quad m_{\text{water}}, \quad m_{\text{ice}}, \quad m_{\text{evap}}$$

NACA0012 – Modeling Choices Across Submissions

NACA0012 – Summary of Submitted Data

AE3932

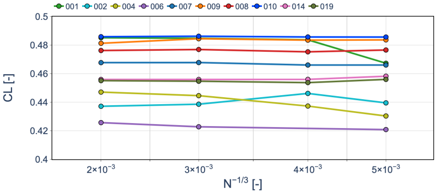
ID	L1 1	L1 3	L1 7	L1 15	L2 1	L2 3	L2 7	L2 15	L3 1	L3 3	L3 7	L3 15	L4 1	L4 3	L4 7	L4 15
001	✓				✓				✓				✓			
002		✓	✓	✓		✓	✓	✓		✓	✓	✓		✓	✓	✓
003					✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓
004	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓
006				✓				✓								✓
007	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓
008		∗				∗				∗				∗		
009		∗				∗				∗				∗		
010	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓
013				✓				✓								
014	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓
015																
019	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓
020				✓				✓	✓	✓	✓	✓				✓

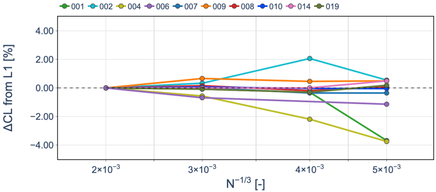
AE3933

ID	L1 1	L1 3	L1 7	L1 15	L2 1	L2 3	L2 7	L2 15	L3 1	L3 3	L3 7	L3 15	L4 1	L4 3	L4 7	L4 15
001	✓	✓	✓	✓	✓				✓				✓			
002		✓	✓	✓		✓	✓	✓		✓	✓	✓		✓	✓	✓
003					✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓
004	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓
006				✓				✓								
007	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓
008	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓
009	✓				✓				✓				✓			
010	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓
013				✓				✓								
014	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓
015																
019	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓	✓
020												✓				

✓: populated CutData received for the corresponding droplet-bin distribution.
∗: Excel CFD results only; no bin count implied.

NACA0012 – Lift Coefficient Grid Convergence

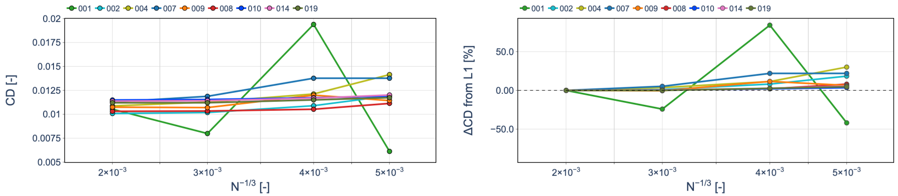




Assessment

- At L1, the median lift coefficient is $C_L = 0.4677$, with an IQR of 0.0262 (5.6%).
- Most submissions exhibit limited sensitivity to spatial grid refinement.
- Relative to L1, the largest coarse-grid deviations are approximately 4%.

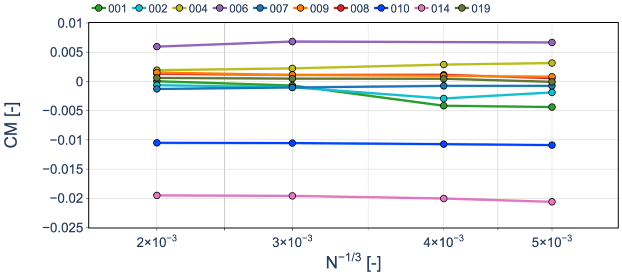
NACA0012 – Drag Coefficient Grid Convergence



Assessment

- At L1, the median drag coefficient is $C_D = 0.01088$, with an IQR of 7.57×10^{-4} (7.0%).
- C_D exhibits greater sensitivity to spatial grid refinement than C_L.
- Non-monotonic grid-convergence behavior is observed for selected submissions.

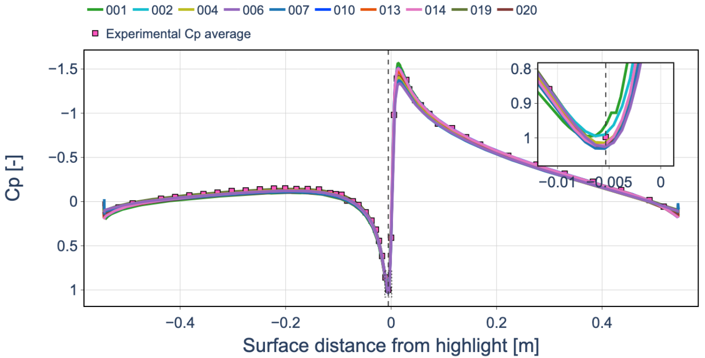
NACA0012 – Pitching Moment Grid Convergence



Assessment

- At L1, the median pitching-moment coefficient is $C_M \approx 0$, with an IQR of 2.60×10^{-3}.
- Most individual submissions exhibit limited sensitivity to spatial grid refinement.
- Code-to-code offsets are larger than the grid-refinement effect for most submissions.
- Relative percentage differences are not meaningful for C_M because the reference values are close to zero.

NACA0012 – Surface Pressure Distribution (L1)



Assessment

- The submitted C_p distributions are closely clustered and show good agreement with the experimental measurements over most of the airfoil.
- The largest code-to-code differences are concentrated near the leading-edge suction peak and stagnation region.
- The overall pressure distribution is consistent across submissions despite differences in prescribed surface roughness.

NACA0012 – Aerodynamic Assessment Summary

Grid Convergence

- C_L: generally weak grid sensitivity, with most submissions remaining close to their L1 prediction.
- C_D: greater grid sensitivity, with non-monotonic behavior observed for selected submissions.
- C_M: generally limited within-code grid sensitivity, while code-to-code offsets remain comparatively large.

Code-to-Code Variability at L1

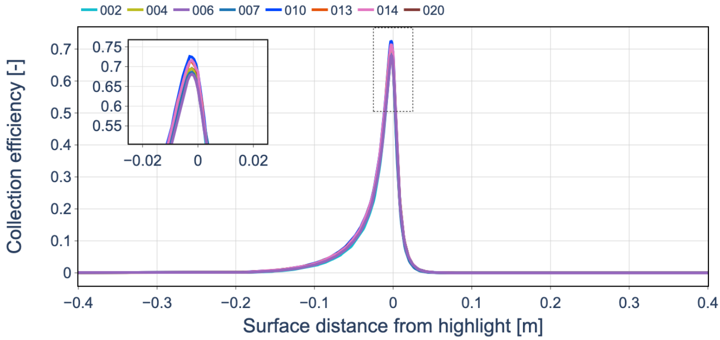
	Median	**IQR**	**Relative IQR**
C_L	0.4677	0.0262	5.6%
C_D	0.01088	7.57×10^{-4}	7.0%
C_M	≈ 0	2.60×10^{-3}	–

Relative IQR is not reported for C_M because normalization by a value close to zero is ill-conditioned.

Aerodynamic Takeaway

The C_p distributions are closely clustered and generally agree well with the experimental measurements. Grid refinement produces relatively small changes in C_L and C_M for most submissions, while C_D is more grid-sensitive. Residual code-to-code differences therefore cannot be attributed to spatial resolution alone.

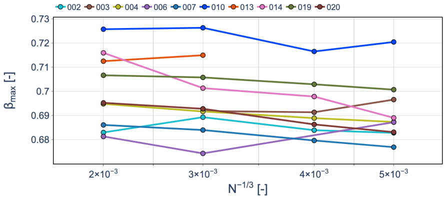
NACA0012 – Collection Efficiency



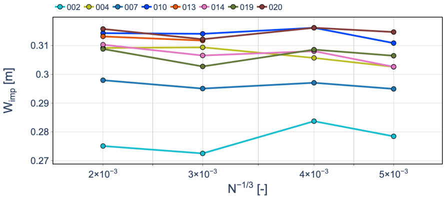
Assessment

- Most submissions predict a similar collection-efficiency distribution around the leading edge.
- Peak collection efficiencies are closely clustered, while differences remain in the detailed profile and impingement extent.
- Differences in prescribed surface roughness should be considered when interpreting the remaining code-to-code variability.

NACA0012 – Collection Efficiency Grid Convergence



Peak Collection Efficiency

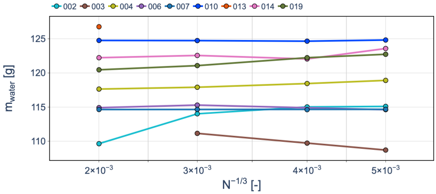


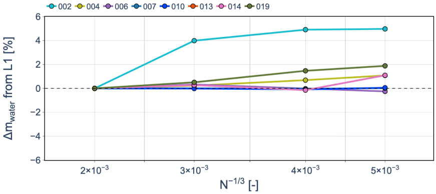
Impingement Width

Assessment

- At L1, the median peak collection efficiency is $\beta_{\text{max}} = 0.695$ with an IQR of 0.0209 (3.0%).
- The L1 median impingement width is 0.309 m with an IQR of 8.93×10^{-3} m (2.9%).
- Both quantities exhibit limited grid sensitivity for most submissions, indicating comparatively stable local impingement predictions.

NACA0012 – Water Mass Grid Convergence

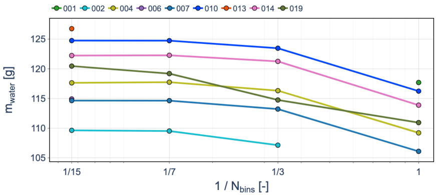


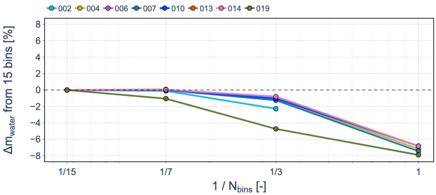


Assessment

- For the 15-bin distribution, the L1 median water mass is 119.05 g with an IQR of 6.39 g (5.37%).
- Most submissions exhibit weak grid sensitivity, with changes generally within approximately 5% of their L1 prediction.
- Code-to-code differences are generally larger than the grid-refinement effect.

NACA0012 – Droplet Distribution Sensitivity





Assessment

- The 15-bin and 7-bin distributions produce very similar impinged water masses.
- Reducing the distribution from 7 to 3 bins introduces only a modest change for most submissions.
- Relative to the 15-bin distribution, the single-bin approximation reduces the collected water mass by approximately 5–8% for most participants.

NACA0012 – Droplet Impingement Summary

Spatial Grid Convergence

- Limited grid sensitivity is observed for most submissions.
- Code-to-code variability generally exceeds the grid-refinement effect.

Droplet-Size Distribution

- The 7- and 15-bin distributions provide very similar results.
- The single-bin approximation predicts approximately 5–8% less collected water.

Icing-Condition Sensitivity

- Little difference in total water impingement is observed between the two icing conditions.

Takeaway

Grid refinement has a limited effect, while droplet-size discretization introduces a more systematic sensitivity. A 7-bin distribution closely reproduces the 15-bin result.

NACA0012 – Convective Heat Transfer Evaluation

Integrated Convective Heat Transfer

The convective heat transfer is evaluated along the surface slice as

$$\boxed{Q_c' = \int HTC(s)\,[T_s(s) - T_{\mathrm{rec}}(s)]\; ds} \qquad [Q_c'] = \mathrm{W/m}.$$

The submitted HTC, surface temperature T_s, and pressure coefficient C_p are used directly.

Recovery Temperature Reconstruction

Since T_{rec} is not directly available from the submissions, it is reconstructed from the submitted C_p distribution:

$$\boxed{C_p(s) \longrightarrow p_e(s) \longrightarrow M_e(s) \longrightarrow T_e(s) \longrightarrow T_{\mathrm{rec}}(s)}$$

Assuming $p_s \simeq p_e$ and an isentropic external flow,

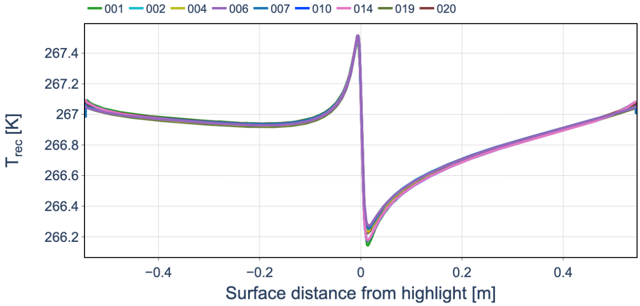
$$\frac{p_e}{p_\infty} = 1 + \frac{\gamma}{2} M_\infty^2\, C_{p,e}.$$

Recovery Model

For a turbulent boundary layer,

$$T_{\mathrm{rec}} = T_e \left[1 + r\,\frac{\gamma - 1}{2} M_e^2\right], \qquad r = Pr^{1/3} \approx 0.9.$$

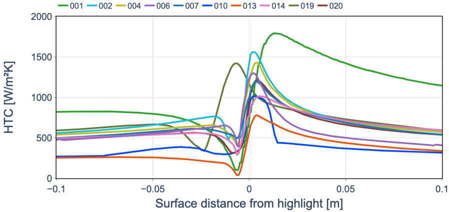
NACA0012 – Recovery Temperature Distribution (L1)



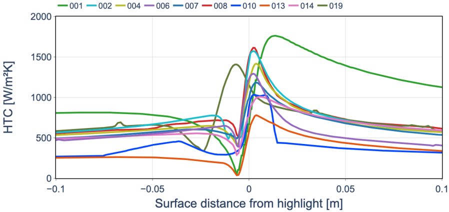
Assessment

- Reconstructed recovery-temperature distributions are closely clustered across submissions.
- This good agreement is consistent with the close agreement previously observed in the C_p distributions.

NACA0012 – Convective Heat-Transfer Coefficient (L1)



AE3932

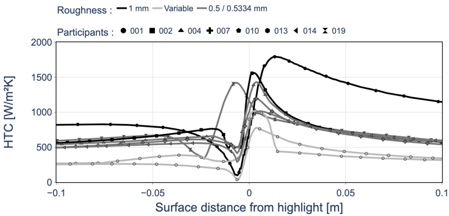


AE3933

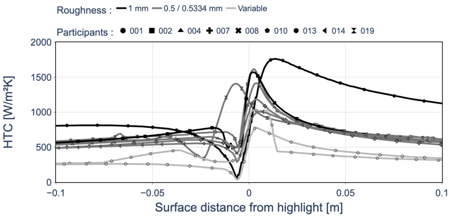
Assessment

- Considerable code-to-code variability is observed in *HTC*, particularly near the leading edge, with similar behavior for both conditions.
- Grouping by prescribed roughness shows that roughness contributes to the observed variability, but does not fully explain the code-to-code differences.

NACA0012 – Convective Heat-Transfer Coefficient (L1)



AE3932 – Grouped by Roughness



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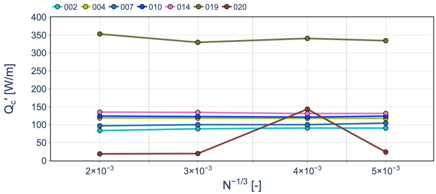
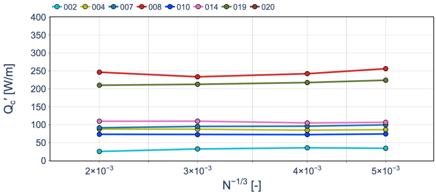
NACA0012 – Convective Heat Transfer Grid Convergence



AE3932

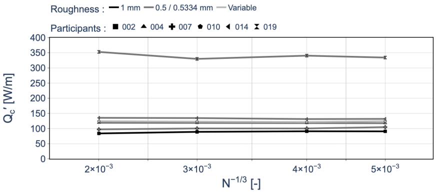


AE3933

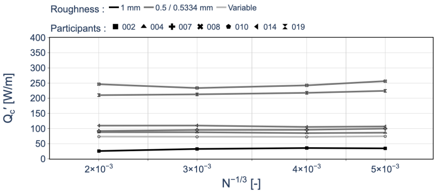
Assessment

- Most submissions exhibit limited grid sensitivity, while code-to-code variability is substantially larger than the grid-refinement effect.
- AE3932 and AE3933 show broadly comparable integrated convective heat-transfer predictions.
- Grouping by prescribed roughness shows a systematic roughness effect, although substantial code-to-code variability remains.

NACA0012 – Convective Heat Transfer Grid Convergence



AE3932 – Grouped by Roughness



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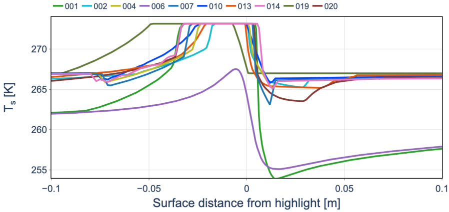
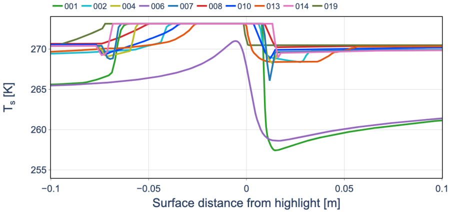
NACA0012 – Surface Temperature Comparison



AE3932

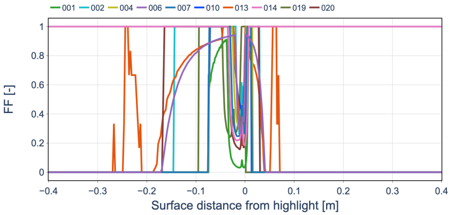


AE3933

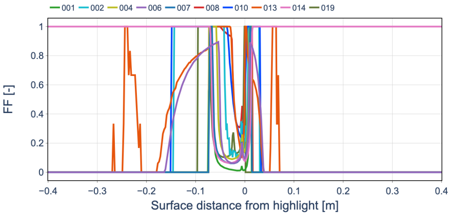
Assessment

- Substantial code-to-code variability is observed in the surface-temperature distributions for both conditions.
- Differences between AE3932 and AE3933 are also evident in the predicted surface thermal response.

NACA0012 – Freezing Fraction Comparison



AE3932



AE3933

Assessment

- Considerable code-to-code variability is observed in the freezing fraction for both conditions.
- Different thermodynamic regimes and transition locations are predicted across submissions.
- Sharp and spatially shifted transitions make direct pointwise comparison particularly difficult.

NACA0012 – HTC & Thermodynamics Assessment

Recovery Temperature

- Closely clustered across submissions, consistent with the good agreement in C_p.

Convective Heat Transfer

- Larger code-to-code variability is observed in *HTC* and Q_c', with roughness explaining only part of the spread.
- Grid sensitivity remains comparatively limited for most submissions.

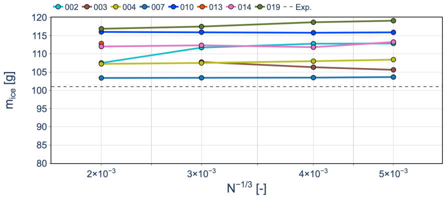
Thermodynamic Response

- Surface temperature and freezing fraction exhibit substantial code-to-code variability.
- Different freezing regimes and transition locations are predicted across submissions.

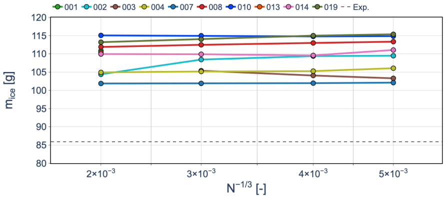
Takeaway

Aerodynamic agreement leads to consistent T_{rec}, while larger differences emerge in *HTC* and propagate into the predicted thermodynamic response.

NACA0012 – Ice Mass Grid Convergence



AE3932

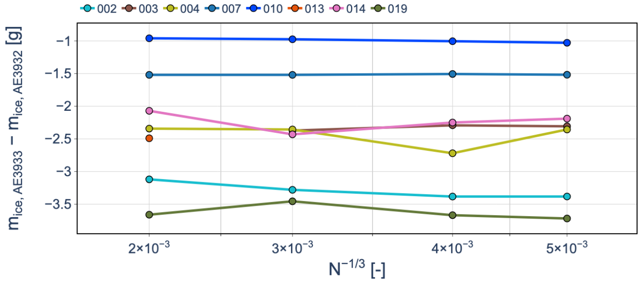


AE3933

Assessment

- Most submissions exhibit limited ice-mass sensitivity to spatial grid refinement for the 15-bin distribution.
- An isolated submission exhibits substantially stronger grid sensitivity than the main participant cluster.
- Code-to-code differences remain visible on the finest grid and cannot be attributed solely to spatial resolution.
- A systematic difference between AE3932 and AE3933 begins to emerge in the predicted ice mass.

NACA0012 – Ice Mass Difference Between Conditions



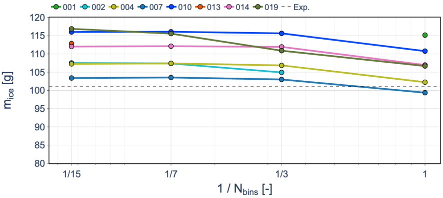
Assessment

- The matched-participant comparison shows lower ice mass for AE3933 than AE3932 for most submissions.
- The difference remains comparatively consistent across the grid hierarchy for most submissions.

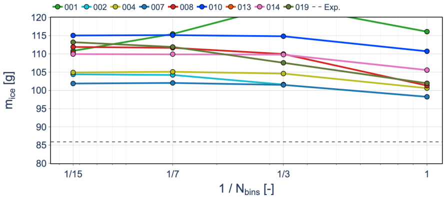
Key Observation

A systematic difference between the two conditions emerges in ice mass; however, the numerical predictions do not reproduce the approximately 15% difference observed experimentally.

NACA0012 – Ice Mass Sensitivity to Droplet Distribution



AE3932

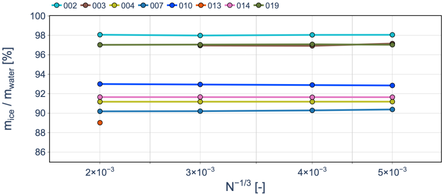


AE3933

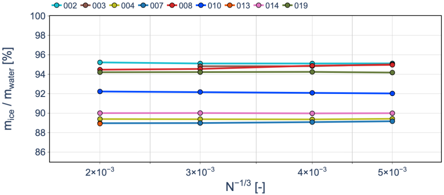
Assessment

- The 15- and 7-bin distributions generally produce similar ice masses.
- Greater sensitivity appears as the droplet distribution is reduced to 3 and particularly 1 bin.
- For most submissions, reducing the droplet distribution modifies the ice mass more noticeably than the 15-to-7-bin reduction.
- The response is participant-dependent, indicating that droplet-size discretization influences the resulting icing response in addition to total water impingement.

NACA0012 – Ice-to-Water Mass Ratio



AE3932

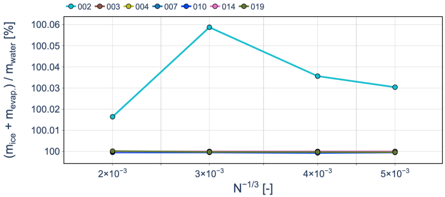


AE3933

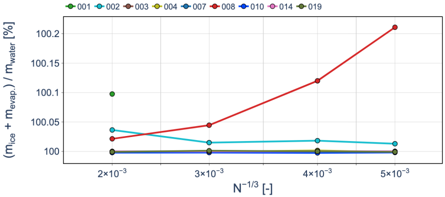
Assessment

- Normalizing by the impinged water mass separates the icing response from differences in total water collection.
- Since the two conditions exhibit comparable water impingement, differences in this ratio directly highlight differences in the resulting ice accretion.
- The comparison therefore provides a clearer measure of the transition from impingement to the thermodynamic icing response.

NACA0012 – Ice and Evaporation-to-Water Mass Ratio



AE3932

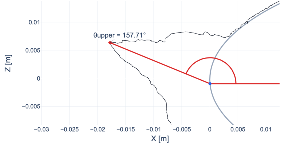


AE3933

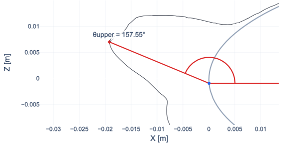
Assessment

- Including evaporated water provides additional information on the partition of the impinged water mass.
- Comparison with $m_{\mathrm{ice}}/m_{\mathrm{water}}$ identifies the contribution of evaporation to the difference between the two icing conditions.
- The remaining fraction represents water leaving the local ice-accretion balance through the other modeled mass-transfer mechanisms.

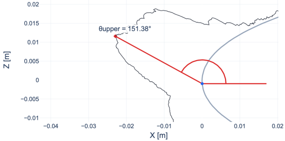
NACA0012 – Experimental Upper Horn Angle



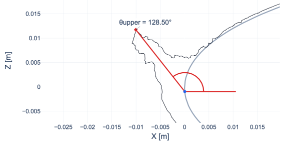
MinCCS



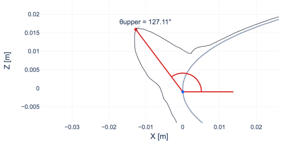
MeanCCS
AE3932



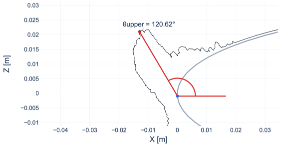
MaxCCS



MinCCS



MeanCCS
AE3933

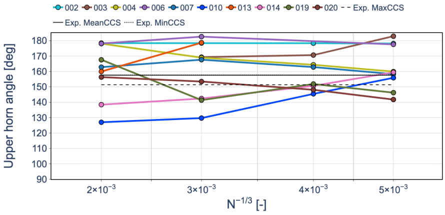


MaxCCS

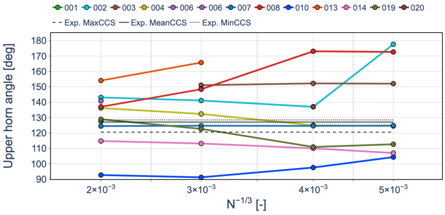
Assessment

- The experimental upper-horn angle is approximately 151–158° for AE3932 and 121–129° for AE3933.
- The experiments therefore indicate a clear change in upper-horn orientation between the two conditions.

NACA0012 – Upper Horn Angle Grid Convergence



AE3932

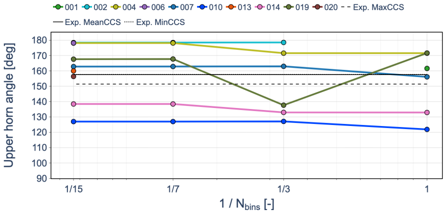


AE3933

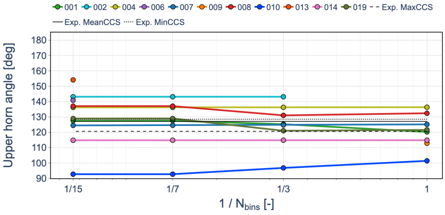
Assessment

- The upper horn angle provides a geometric grid-convergence metric complementary to the integrated ice mass.
- Differences in horn orientation quantify changes in where the accreted ice is distributed rather than only how much ice is formed.
- Comparison between AE3932 and AE3933 indicates whether the change in icing response also produces a systematic change in horn orientation.

NACA0012 – Upper Horn Angle Sensitivity to Droplet Distribution



AE3932

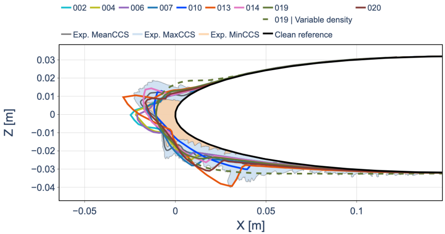


AE3933

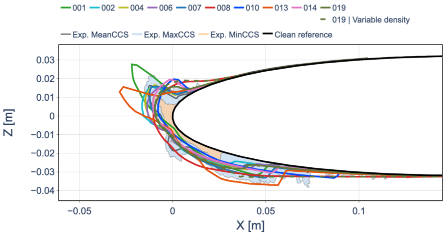
Assessment

- The L1 comparison evaluates the sensitivity of the upper horn orientation to droplet-size discretization.
- This provides a geometric counterpart to the ice-mass droplet-distribution sensitivity.
- Changes in total ice mass and changes in horn orientation should therefore be considered together when assessing droplet-distribution convergence.

NACA0012 – Ice Shapes (L1, 15 Bins)



AE3932

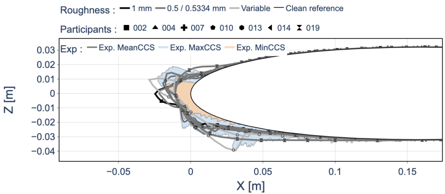


AE3933

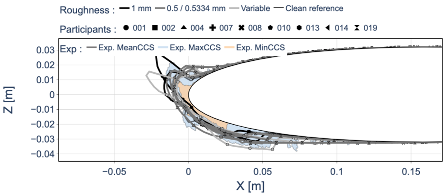
Assessment

- Considerable code-to-code variability is observed in the predicted ice shapes for both conditions.
- Experimental ice shapes provide the validation reference for the predicted accretion geometry.
- Grouping by prescribed roughness helps assess the contribution of roughness to the observed geometric variability.

NACA0012 – Ice Shapes (L1, 15 Bins)



AE3932 – Grouped by Roughness

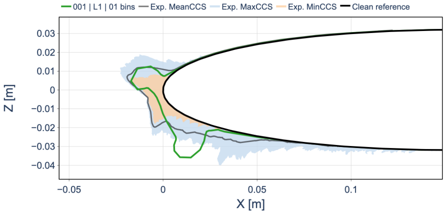


AE3933 – Grouped by Roughness

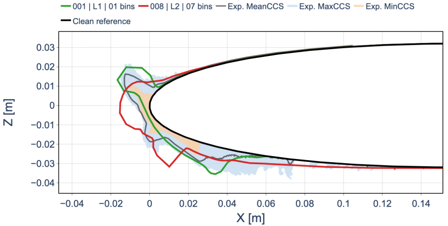
Assessment

- Considerable code-to-code variability is observed in the predicted ice shapes for both conditions.
- Experimental ice shapes provide the validation reference for the predicted accretion geometry.
- Grouping by prescribed roughness helps assess the contribution of roughness to the observed geometric variability.

NACA0012 – Multilayer Ice-Shape Comparison (L1)



AE3932



AE3933

Assessment

- The multilayer predictions are compared directly with the experimental ice-shape envelopes for both conditions.
- Differences are observed in the predicted horn extent, orientation, and local ice thickness.
- The multilayer comparison provides the final assessment of the evolving ice-accretion methodology.

NACA0012 – Ice Accretion Summary

Ice Mass

- Most submissions show limited grid sensitivity, while a systematic difference emerges between AE3932 and AE3933.

Mass Partition

- Mass ratios highlight differences in freezing and evaporation despite comparable water impingement.

Ice Geometry

- Horn metrics and ice shapes characterize how differences in ice mass translate into the final accretion geometry.

Takeaway

While aerodynamics, impingement, and heat transfer remain broadly comparable, the resulting ice accretion distinguishes AE3932 and AE3933.

NACA0012 – Overall Assessment

Aerodynamics

- Limited grid sensitivity overall, with larger sensitivity in C_D; L1 C_p agrees well with experiment.

Droplet Impingement

- Limited grid sensitivity and similar impingement between conditions; droplet-size discretization has a more systematic effect.

Convective Heat Transfer

- Recovery temperature is closely clustered, while larger code-to-code variability is observed in HTC.

Ice Accretion

- The clearest distinction between AE3932 and AE3933 emerges in the resulting ice accretion.